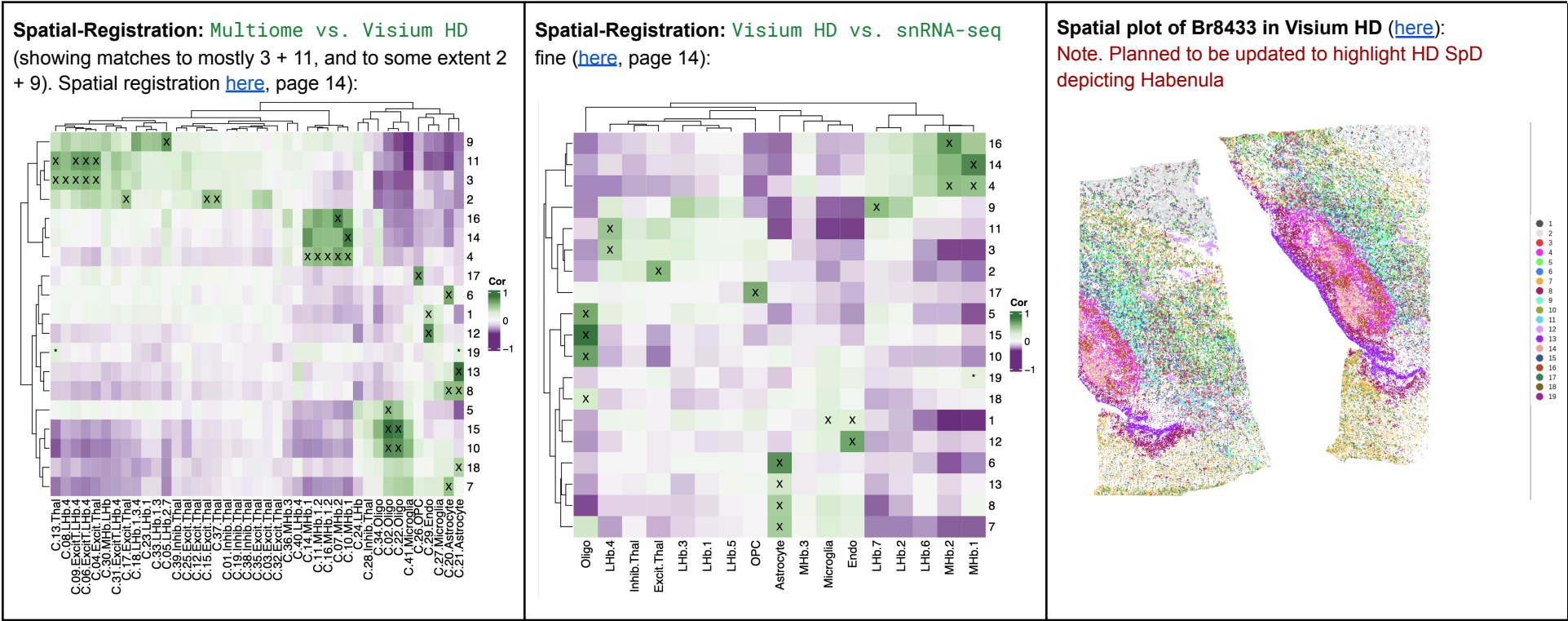


# Resolving Habenula Multiome WNN Ambiguous Clusters with Visium-HD

Cynthia SC, Nick Eagles ( July, 2025)

The **WNN multiome clusters** below were labeled as **ambiguous** because spatial registration against scRNAseq and/or Visium'3, mean-ratio, and top 50 gene frequency analyses didn't provide consistent support for a unique cell type. To clarify, we used the higher resolution of the **Habenula Visium-HD** data, together with **spot-plots** depicting the Hb expected region. Below, we summarize the findings and the technical evidence behind our final cell-type assignments.

| WNN cluster                  | Cell-type label | Technical support                                          |
|------------------------------|-----------------|------------------------------------------------------------|
| C.04, C.06, C.09, C.13, C.31 | *LHb.4          | VisiumHD resolution 1.4 - 3 + 11, and to some extent 2 + 9 |





| WNN cluster | Cell-type label | Technical support                              |
|-------------|-----------------|------------------------------------------------|
| C.30        | *LHb.4          | VisiumHD resolution 0.4 for LHb<br>C.30 SpD=02 |
| C.34        | *Oligo          | C.34 SpD=04                                    |

